CENTENARYO: A Web-Based Centralized End-to-end Network for Transactional Entitlement, Notification, Anomaly Review, and Year-end Oversight using Random Forest Anomaly Detection and Prescriptive Audit Analytics.

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I. Introduction

In many Local Government Units (LGUs), the Office of the Senior Citizens Affairs (OSCA) serves as the primary gateway for elderly citizens to access social services. However, as populations grow, traditional methods of record-keeping—such as fragmented Excel sheets and paper-based ledgers—have become major bottlenecks. These manual systems are not only slow but are also susceptible to "ghost" pensioner claims, where benefits are issued to deceased or ineligible individuals due to a lack of real-time data synchronization.

To address these systemic inefficiencies, this study proposes **CENTENARYO** (A Web-Based Centralized End-to-end Network for Transactional Entitlement, Notification, Anomaly Review, and Year-end Oversight). Built on a robust PHP/Python and MySQL stack, the system centralizes all records into a single on-premise relational database. Unlike traditional databases, **CENTENARYO** integrates *Random Forest Anomaly Detection* to proactively identify suspicious records and provides *Prescriptive Audit Analytics* to guide LGU decision-makers. Deployed on the local City Hall network, it ensures a secure, high-speed environment that remains fully operational without the need for a stable internet connection.

I.A. Statement of the Problem

The transition toward efficient social welfare delivery for the elderly is currently hindered by the following critical challenges in OSCA's manual operations:

- **Proliferation of "Ghost" Pensioners and Fraud:** The lack of an intelligent validation system means benefits are often inadvertently released to deceased individuals or duplicate identities, leading to significant fiscal leakages.
- **Inefficient Manual Payrolls:** Creating quarterly disbursement lists requires days of manual cross-referencing across multiple files, a process highly prone to human error and data entry fatigue.
- **Poor Inventory Accountability:** Medicine and grocery discount booklets are tracked via paper logs, making it nearly impossible to prevent duplicate issuances or verify the history of replaced items in real-time.

- **Lack of Prescriptive Insight:** Current systems only provide basic summaries. There is no automated mechanism to "flag" anomalies or predict future budgetary needs based on demographic shifts and mortality trends.

I.B. Objectives of the Study

The primary objective of this project is to develop and deploy **CENTENARYO**, an AI-enhanced, secure, and offline-capable system that automates OSCA functions while utilizing Random Forest algorithms to ensure the integrity of social pension disbursements.

I.B.i. Specific Objectives

1. **To Centralize Record Management:** Develop a master profile registry with unique identifiers to eliminate duplicate registrations and serve as the "Single Source of Truth."

2. **To Automate Disbursement Workflows:** Create a "one-click" payroll generator that cross-references active records with vital status updates for error-free disbursement.

3. **To Implement AI-Driven Anomaly Detection:** Utilize a *Random Forest algorithm* to analyze transaction patterns and flag high-risk accounts for further audit and verification.

4. **To Improve Inventory Tracking:** Implement a digital serialization system for discount booklets and a printing module for standardized Senior IDs to ensure full accountability of physical assets.

5. **To Enable Prescriptive Audit Analytics:** Design a dashboard that provides data-driven recommendations for budget allocation and identifies vulnerable sub-populations requiring immediate intervention.

6. **To Ensure Operational Security:** Deploy the system on a Local Area Network (LAN) with comprehensive audit logs to record every staff transaction for transparency and accountability.